TEXORA · LECTURE SHEET

Production Systems, Capacity Planning and the Functions of PPC

What a factory produces, how much it can produce, and how production is planned and controlled — production types and life cycle, capacity planning, process planning and design, the PPC functions from forecasting to dispatching, plant layout and location, and make-or-buy decisions.

TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 1 · LEVEL 4 TERM 1

1. Functions and types of production

Production is the transformation of inputs — materials, labour, machine time, energy, information — into outputs of greater value. Its functions in a textile enterprise are: converting raw material through successive processes (spinning → weaving/knitting → wet processing → garmenting), holding quality at the specified standard at every stage, delivering on the promised date, and doing all of this at a cost the price can bear. Production systems are classified by volume and variety:

TYPE	VOLUME / VARIETY	LAYOUT LOGIC	TEXTILE EXAMPLE
Job production	One-off or tiny lots, high variety	General-purpose machines, skilled labour	Sample room; a single specialised technical-textile order
Batch production	Medium lots repeated irregularly	Process layout, changeovers between lots	Dye house dyeing many colours in lots; a weaving shed running many fabric styles
Mass / flow production	High volume, low variety	Product layout, dedicated lines	Spinning mill producing one count for months; denim plant
Continuous production	Uninterrupted flow, material transformation	Fixed plant, automatic control	Nonwoven line, continuous dyeing/finishing range

Most composite textile mills are hybrids: continuous at the finishing range, batch in dyeing, mass in spinning, job work in the sample section. PPC must plan each zone in its own logic — a single planning method imposed on all four is a common cause of chaos.

2. The production (product) life cycle

A product passes through **introduction** (low volume, high unit cost, frequent design change), **growth** (rising orders, capacity strain), **maturity** (stable volume, price pressure, cost competition) and **decline** (falling demand, spare capacity). The production implication changes with the stage: introduction needs flexibility and fast sampling; growth needs capacity added ahead of demand; maturity needs cost and efficiency work — work study, waste control; decline needs the discipline to release capacity to the next product rather than defend a dying one. In garments this cycle may run in a single season; in commodity yarn it may run for years.

3. Capacity planning

Capacity is the maximum rate of output a facility can sustain under realistic conditions. Three measures must not be confused: **design (rated) capacity** — the nameplate rate; **effective capacity** — the rate achievable after allowances for changeovers, maintenance, breaks and expected efficiency; and **actual output**. Two ratios follow:

$$\text{Utilisation} = \text{Actual output} \div \text{Design capacity} \quad | \quad \text{Efficiency} = \text{Actual output} \div \text{Effective capacity}$$

WORKED EXAMPLE — WEAVING SHED CAPACITY

40 looms, rated output 25 m/h each, working 2 shifts × 8 h for 26 days a month.

Design capacity = 40 × 25 × 16 × 26 = **416,000 m/month**.

Actual output recorded: 340,000 m. Utilisation = 340,000 ÷ 416,000 × 100 = **81.7 %**.

Reading: nearly a fifth of the installed capacity produced nothing — the planner's task is to find where (order gaps? beam-gaiting queues? style changeovers?) before recommending new looms.

Capacity decisions are long-lead and lumpy: a dyeing machine or a boiler cannot be bought in weekly increments. Planners therefore work with a **capacity cushion** (spare capacity against demand surges) whose size depends on demand variability and the cost of lost orders: fashion garmenting, with violent seasonal peaks, needs a larger cushion — or subcontracting arrangements that act as hired cushion — than a commodity yarn mill.

4. Process planning and process design

Process planning converts a product specification into a manufacturing route: which operations, in what sequence, on which machines, with what settings and what standard times. Its document in textiles is the route/process sheet — for a fabric: yarn count and construction → sizing recipe → loom type and settings → grey inspection standard → dyeing/finishing route → final inspection. **Process**

design goes one level deeper and fixes the process variables themselves (temperatures, speeds, chemical doses) — the work that TE-3207 and WP-4101 style calculations feed into. Good process planning shows up as few surprises on the floor; poor planning shows up as departments each optimising themselves against the others.

5. The functions of PPC

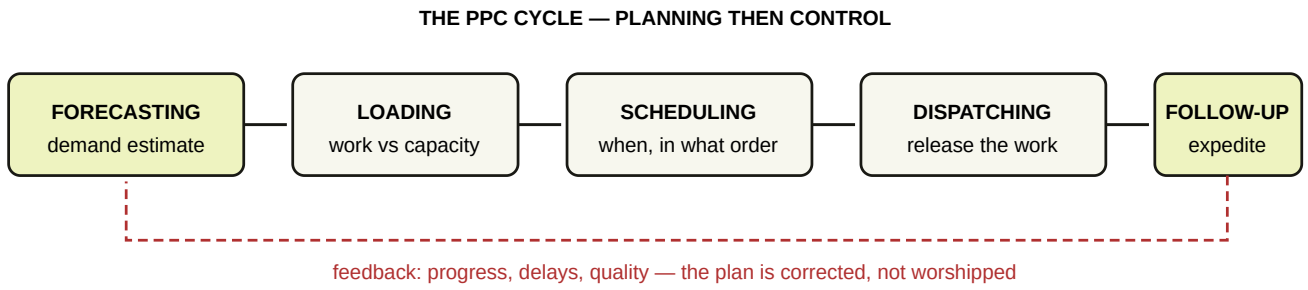


Fig. 1 — The PPC functions as a cycle: the first four functions plan, dispatching executes, follow-up feeds reality back into the next plan.

- **Forecasting** — estimating future demand from order books, buyer programmes, seasonality and market intelligence; in garments, forecasts are per style/colour/size ratio and are revised as bookings firm up.
- **Loading** — comparing the work demanded with capacity machine-hour by machine-hour and department by department, exposing overloads (overtime, subcontract, reschedule) and underloads (bring work forward, seek orders) before they happen.
- **Scheduling** — fixing start and finish times and sequence for each job on each resource; master schedule at mill level, detailed schedules at department level; Gantt charts and line plans are its working tools.
- **Dispatching** — the point where planning becomes instruction: issuing job cards, material requisitions, tool and programme releases so that work actually starts.
- **Follow-up / expediting** — tracking progress against the schedule, identifying delays and their causes, and replanning. Without follow-up the other four functions are decoration.

6. Plant layout and plant location

Layout arranges departments and machines to minimise material travel, backtracking and work-in-progress while keeping services (steam, water, effluent, humidification) short and safe. The main patterns are **process layout** (machines grouped by function — flexible, suited to batch production), **product/line layout** (sequence follows the product — fast flow, suited to mass production), and **fixed-position layout** (product stays, resources come to it — rare in textiles except in plant installation itself).

Textile flow is heavy and wet in the middle: a sound composite layout runs spinning → weaving → wet processing → finishing in a straight or U flow with the effluent plant at the wet end and the boiler house central to steam users.

Location decisions weigh: raw material and market access, labour supply and skill, power/water reliability, effluent disposal permission, land cost, transport links and incentives (e.g. export processing zones). Location errors are nearly permanent — a dye house built where effluent treatment is impossible has bought a lifetime problem.

7. Purchasing principles and make-or-buy

Purchasing for production follows five principles: the **right quality** (specified, not gold-plated), **right quantity** (economic order quantities balanced against storage and working capital), **right time** (aligned to the production schedule — early delivery is also a cost), **right price** (total cost of ownership, not invoice price alone) and **right source** (reliable suppliers, rated as in TE-4108). The sharpest purchasing decision is **make-or-buy**: produce a component or service in-house, or buy it? The comparison must use *avoidable* costs only — in-house variable cost plus any fixed cost that would genuinely change:

> **WORKED EXAMPLE — MAKE OR BUY SEWING-THREAD CONES WINDING**
>
> A garment factory needs 20,000 wound cones a year. In-house: variable cost Tk 42 per cone; avoidable fixed cost (a winder's depreciation and supervision that would disappear if bought) Tk 1,80,000.
>
> In-house unit cost = (20,000 × 42 + 180,000) ÷ 20,000 = **Tk 51 per cone**.
>
> Supplier's price: Tk 48 per cone → **buy**, saving (51 − 48) × 20,000 = **Tk 60,000 a year** — provided quality, shade matching and lead time are equal. Fixed costs that would continue either way (factory rent, the PPC office) are *excluded*: including them is the classic make-or-buy error.

Glossary

Capacity — maximum sustainable output rate. **Design capacity** — nameplate rate. **Effective capacity** — rate realistically achievable after allowances. **Utilisation** — actual output ÷ design capacity. **Loading** — assigning work to resources and comparing with capacity. **Scheduling** — timing and sequencing jobs. **Dispatching** — releasing work to the floor. **Capacity cushion** — spare capacity held against demand variation. **Avoidable cost** — a cost that disappears if an activity stops; the only fixed cost relevant to make-or-buy.

Key points

• Textile mills mix job, batch, mass and continuous production — PPC must respect each zone's logic.
• Product life-cycle stage dictates whether flexibility, capacity, cost or exit discipline matters most.
• Design capacity 416,000 m/month vs actual 340,000 m gives 81.7 % utilisation in the worked example — find the lost fifth before buying machines.
• PPC cycle: forecasting → loading → scheduling → dispatching → follow-up, with feedback correcting the plan.
• Make-or-buy compares avoidable cost only: Tk 51 in-house vs Tk 48 bought — buy, saving Tk 60,000 a year.

Practice

1. A knitting section has 24 machines rated at 18 kg/day each, working 25 days a month, and produces 8,900 kg in the month. Compute design capacity and utilisation. *(Hint: capacity = 24 × 18 × 25.)*
2. Distinguish utilisation from efficiency of capacity, and say which one a sales manager and which one a production manager should watch.
3. A component costs Tk 30 variable + Tk 90,000 avoidable fixed for 10,000 units in-house; a supplier quotes Tk 36. Decide, with figures.
4. An order book suddenly doubles for one month. Using the PPC functions in order, list what the planner does in the first week.

Answer key

1. Design capacity = 24 × 18 × 25 = 10,800 kg. Utilisation = 8,900 ÷ 10,800 × 100 ≈ 82.4 %.
2. Utilisation = actual ÷ design capacity — it measures how much of the investment is being sweated, the sales/commercial concern (are orders filling the plant?). Efficiency = actual ÷ effective capacity — it measures how well the plant performs against what is realistically achievable after allowances, the production manager's concern. A plant can show low utilisation (weak order book) with high efficiency (well run whenever it runs).
3. In-house unit cost = (10,000 × 30 + 90,000) ÷ 10,000 = Tk 39. Supplier Tk 36 → buy, saving (39 − 36) × 10,000 = Tk 30,000, subject to quality and lead-time equivalence.
4. Forecasting: confirm which orders are firm and their delivery dates; Loading: total the machine-hours demanded per department against capacity — identify the bottleneck department; Scheduling: resequence jobs, plan overtime/subcontract for the bottleneck, protect confirmed export dates; Dispatching: release only work whose materials are available, in the new sequence; Follow-up: daily progress checks against the revised schedule, expediting late materials, and immediate replanning when a job slips.